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RESEARCH ARTICLE
10.1029/2021JB022938

Special Section:
Understanding and anticipating Induced Seismicity: from mechanics to seismology

Temporal Changes in Seismic Velocity and Attenuation at The Geysers Geothermal Field, California, From Double-Difference Tomography

Hao Guo[1] **and Clifford Thurber**[1]

[1]Department of Geoscience, University of Wisconsin-Madison, Madison, WI, USA

Key Points:

- High-resolution imaging of temporal changes in Vp and Qp reveals detailed injection-induced changes in fracturing and saturation conditions
- Decreased Vp and Qp and diffuse seismicity suggest diffuse fluid flow, increased fracturing, and increased but still partial saturation
- Decreased Vp, increased Qp, and concentrated seismicity suggest confined fluid flow, increased fracturing, and full saturation

Supporting Information:

Supporting Information may be found in the online version of this article.

Correspondence to:

H. Guo,
hao.guo@wisc.edu

Citation:

Guo, H., & Thurber, C. (2022). Temporal changes in seismic velocity and attenuation at The Geysers geothermal field, California, from double-difference tomography. *Journal of Geophysical Research: Solid Earth*, *127*, e2021JB022938. https://doi.org/10.1029/2021JB022938

Received 1 AUG 2021
Accepted 17 APR 2022

Author Contributions:

Conceptualization: Hao Guo, Clifford Thurber
Data curation: Hao Guo
Formal analysis: Hao Guo
Funding acquisition: Clifford Thurber
Investigation: Hao Guo
Methodology: Hao Guo, Clifford Thurber

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Abstract Water injection and Enhanced Geothermal System (EGS) technologies have been used to exploit heat resources from geothermal reservoirs. Detecting spatial and temporal changes in reservoir physical properties is important for monitoring reservoir condition changes due to water injection and EGS. Here, we determine high-resolution models of the temporal changes in the three-dimensional *P* wave velocity and attenuation (Vp and Qp) structures between the years 2005 and 2011 in the northwestern part of The Geysers geothermal field, California, using double-difference seismic velocity and attenuation tomography. The northwest Geysers has a shallow normal temperature reservoir (NTR) underlain by a high temperature reservoir (HTR) that has substantial underutilized heat resources but may be more fully utilized in the future through EGS. In the southeastern part of the northwest Geysers, however, EGS has been successfully but unintentionally applied for at least 50 years because the waters injected into the NTR have been flowing into the HTR. Our models are well resolved in this area and show that the NTR and HTR have different seismic responses (seismicity, Vp, and Qp) to water injection, which can be explained by the injection-induced differences in fracturing and saturation that are likely related to their geological properties. Our results indicate that the joint analysis of changes in seismicity, velocity, and attenuation is valuable for characterizing changes in reservoir fracturing and saturation conditions. Our results suggest that high-permeability zones and/or pre-existing permeable fault zones are important for the success of EGS at The Geysers and potentially other geothermal systems.

Plain Language Summary To deal with the declining steam pressure and fluid content, water injection is a commonly employed technology to promote the production of geothermal resources from the naturally permeable geothermal reservoirs. To exploit heat energy from the hot, dry but impermeable rocks, water injection technology can be used to create permeability and increase fluid saturation. Such a man-made reservoir is called Enhanced Geothermal System (EGS). Monitoring changes in material properties of reservoirs due to water injection and EGS is important for evaluating and adjusting the production and injection strategies. Here, we use seismic tomography to image temporal changes in the three-dimensional subsurface seismic velocity and attenuation structures of the northwestern part of The Geysers geothermal field, California. The northwest Geysers has a normal temperature reservoir (NTR) at ~1–2-km depths underlain by a high temperature reservoir (HTR). The southeastern part of the northwest Geysers has been exploited for >50 years with unintentionally created but successful EGS. Our joint analysis of changes in seismicity and seismic velocity and attenuation clearly reveal different injection-induced changes in fracturing and saturation conditions in the NTR and HTR in this area that are likely related to their geological properties.

1. Introduction

The Geysers, a vapor-dominated geothermal field in northern California, is the largest geothermal electricity generating operation in the world and also has the most active seismicity in northern California (Majer & Peterson, 2007). It lies within the San Andreas Fault system and is subject to tectonic forces due to the strike slip motion between the North American and Pacific plates (Figure 1). Its shallow reservoir, at depths of ~1–2 km below sea level, has a temperature of ~240 °C and is formed by fractured and hydrothermally altered metagraywacke, which is overlain by a caprock formed by greenstone and unfractured metagraywacke (Garcia et al., 2016). Below this normal temperature reservoir (NTR), there is a high temperature reservoir (HTR, ~300–400 °C) of low permeability in the northwest Geysers with lateral variations in the depth to its top from ~1.8 to 2.2 km below sea

Project Administration: Clifford Thurber
Resources: Clifford Thurber
Software: Hao Guo
Supervision: Clifford Thurber
Validation: Hao Guo, Clifford Thurber
Visualization: Hao Guo
Writing – original draft: Hao Guo
Writing – review & editing: Hao Guo, Clifford Thurber

level (Figure 1), but other than temperature it has no lithologic or mineralogic differences from the NTR (Garcia et al., 2016; Walters et al., 1988).

For The Geysers, water injection since the 1970s has been important for steam production from the NTR, especially in recent decades. Without water injection, the reservoir pressures and steam flow rates would decline rapidly to uneconomical levels (Majer & Peterson, 2007). The largest injection project at The Geysers in the last two decades is the Santa Rosa-Geysers Recharge Project (SRGRP; Figure 1), which has been injecting large volumes of tertiary-treated municipal wastewater from Santa Rosa into the geothermal reservoir since November 2003 (Stark et al., 2005).

Compared to the NTR, the HTR in the northwestern part of The Geysers is underutilized due to low permeability, high concentrations of noncondensable gas, and corrosive chloride (Garcia et al., 2016; Majer & Peterson, 2007). In the southeastern part of the northwest Geysers where most SRGRP wells are located (Figure 1), however, significant volumes of the water injected into the NTR have been flowing downward into the HTR since 1967 or earlier. This is signified by the existence of deep-induced seismicity and the strong temporal correlation between injection volumes and the count of deep-induced earthquakes and may have contributed significantly to the productivity of this part of the field (Stark, 2003). Although not intentional, this is very similar to the Enhanced Geothermal System (EGS) concept, which is to inject cold water into hot, dry, and low-permeability zones to stimulate permeability in order to exploit the heat resources, as has been done by an EGS Demonstration Project conducted in the northwest Geysers (Garcia et al., 2016; Rutqvist et al., 2016). Thus, this area provides an excellent and valuable opportunity to study how water injection induces short-term and long-term changes of both the high-permeability and low-permeability geothermal reservoirs.

Understanding the temporal changes of the NTR and HTR reservoir conditions due to water injection, such as changes in porosity, permeability, saturation, and/or fluid pressure, is important for evaluating and potentially adjusting the production and injection strategies. The most common way to achieve this is through induced seismicity monitoring. Injection-induced seismicity at The Geysers is hypothesized to be caused by cooling-induced rock contraction and reactivation of pre-existing fractures due to increased fluid pressure (Guilhem et al., 2014; Majer & Peterson, 2007; Martínez-Garzón et al., 2014, 2016; Ross et al., 1996; Rutqvist & Oldenburg, 2008). Thus, seismicity can be used to track fluid paths and characterize newly created or pre-existing fracture networks. Moreover, many source characteristics of induced seismicity, e.g., the spatial distance to wellhead, moment magnitude, faulting regime, earthquake stress drop, and volume of the cluster, and the local stress tensor orientation in which the induced earthquakes occur, have been found to be correlated with long-term and/or short-term changes in fluid injection rates by detailed studies on an induced seismicity cluster in the northwest Geysers (Kwiatek et al., 2015; Martínez-Garzón et al., 2013, 2014; Staszek et al., 2017). Thus, analyzing source characteristics and the stress field can contribute to characterize spatiotemporal changes in reservoir conditions due to fluid injection.

Seismic body-wave velocity tomography has also been shown to be effective for detecting spatiotemporal changes of reservoir properties at The Geysers (Foulger et al., 1997; Gunasekera et al., 2003; Hutchings et al., 2014). Seismic attenuation is expected to be more sensitive to reservoir property changes than seismic velocity based on lab experiments (Winkler & Nur, 1979), but its determination is much more challenging due to the greater complexity in making attenuation measurements compared to measuring body-wave arrival times. Several three-dimensional (3-D) attenuation tomography studies have been performed for different parts of The Geysers (Guo & Thurber, 2021; Romero et al., 1997; Zucca et al., 1994). To our knowledge, the only four-dimensional (4-D) attenuation tomography study that imaged temporal changes in attenuation structure at The Geysers is by Hutchings et al. (2014), who imaged short-term (∼1–2 months) attenuation change in an EGS Demonstration Project area.

In this paper, we use the double-difference velocity (DDV) tomography method (Zhang & Thurber, 2003, 2006) and the recently developed double-difference attenuation (DDQ) tomography method (Guo & Thurber, 2021) to determine high-precision earthquake locations and high-resolution *P* wave velocity and attenuation (Vp and Qp) models for The Geysers for the years 2005 and 2011. The inverted models are used to determine temporal changes of Vp and Qp structure (dVp and dQp) between the two periods. It is noted that Guo and Thurber (2021) previously determined the 2011 Vp and Qp models of The Geysers using the DDV and DDQ tomography methods. Compared to Guo and Thurber (2021), the major new achievements of this study include the determination of the 2005 Vp and Qp models and the temporal change models between 2005 and 2011; the 2011 Vp and Qp

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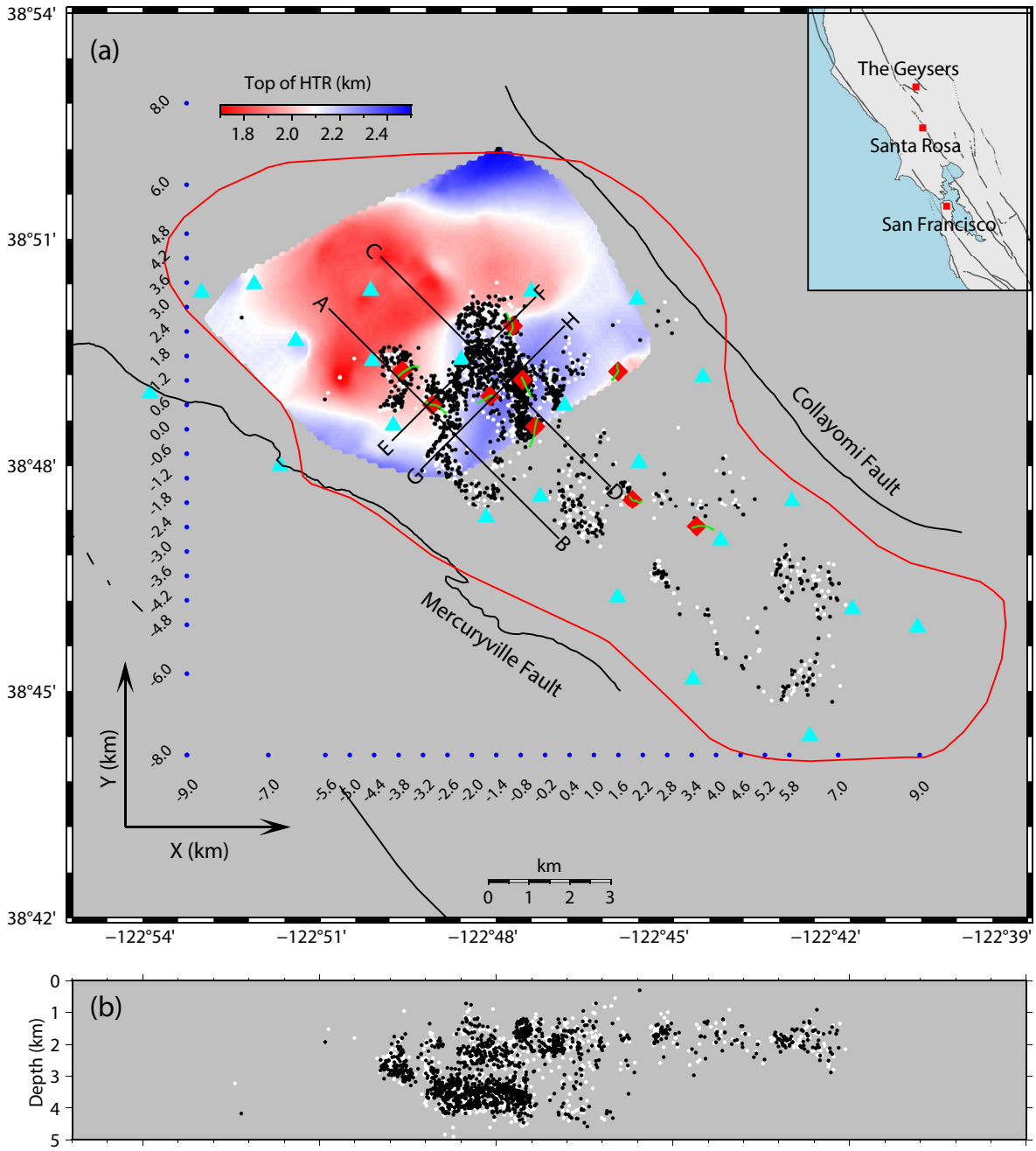


Figure 1. (a) Map view of the earthquakes (black and white dots) and seismic stations (cyan triangles) used for tomographic inversions in this study, and trajectories (short green lines) of Santa Rosa-Geysers Recharge Project (SRGRP) injection wells (red diamonds) at The Geysers geothermal field (outlined by the red line). The colored area in the northwest Geysers maps the top depth of the high temperature reservoir (HTR). White and black dots represent the relocations of the 2005 and 2011 earthquakes used in this study, respectively. Cyan triangles represent the stations with common *P* wave arrival picks and common *P* wave t^* measurements for the selected pairs of 2005 and 2011 earthquakes. Nodes used for the tomographic inversion are shown as blue dots with their coordinates labeled. Two regional faults, Collayomi and Mercuryville, are shown. Lines AB-GH show the locations of the cross-sections shown in Figures 3–5. The inset map at the top right shows the location of The Geysers and the main California fault system (lines). (b) East-west cross-section of the relocations of the 2005 (white dots) and 2011 (black dots) earthquakes used in this study.

models are also updated. The temporal change models are then used to understand the injection-induced temporal changes of the NTR and HTR reservoir conditions between 2005 and 2011 beneath the southeastern part of the northwest Geysers where our models are best resolved (Figure 1). Although there are many active injection wells

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during our study period, our discussions will focus on the ones with large injection volumes in our well resolved area, most of which are SRGRP wells (Figure 2).

2. Data and Methods

2.1. Data

The current seismic network at The Geysers has been operated since 2003 and has been upgraded over time (Gritto & Jarpe, 2014; Majer & Peterson, 2007). There were 26 stations in 2005, which increased to 33 with extension of coverage to the north in 2011 (Figure S1 in Supporting Information S1). Most stations are from the Berkeley Geysers Network and have been operated by Lawrence Berkeley National Laboratory, with two (GAXB and GDXB) from the Northern California seismic network. The processing of seismicity, including earthquake detection, arrival picking, and estimating earthquake location and magnitude, has been done automatically as described in Majer and Peterson (2007). This information and seismic waveform data are stored at the Northern California Earthquake Data Center (NCEDC, 2014). Gritto and Jarpe (2014) relocated 125,000 earthquakes that occurred between 2004 and 2011 using a version of the SIMUL code (Thurber, 1983), which jointly inverts for earthquake locations and velocity model. They then constructed a high-quality catalog for 20,000 events during this period, requiring at least 16 phase picks for each earthquake and a maximum root-mean-square (RMS) residual of 0.05 s for the associated arrival picks. We selected earthquakes in 2005 and 2011 from their high-quality catalog for this study, maximizing the length of the time interval with 23 common stations from the Berkeley Geysers Network (Figure 1). Details about our earthquake selection process are described in Section 3.1.

2.2. DDV and DDQ Tomography Methods

We used the DDV tomography method (Zhang & Thurber, 2003, 2006) and the recently developed DDQ tomography method (Guo & Thurber, 2021) to image the Vp and Qp structures of The Geysers.

DDV tomography uses absolute arrival times from events to stations and differential times from pairs of events at common stations to invert for earthquake locations and velocity models simultaneously. The differential times can be constructed by differencing the absolute times, termed catalog differential times, and/or by waveform cross-correlation (WCC). DDV tomography is able to determine higher-precision relative locations between events and higher-resolution source-region velocity structure compared to conventional velocity tomography (Zhang & Thurber, 2003, 2006). See Text S1 in Supporting Information S1 for details on the methodology of DDV tomography.

DDQ tomography involves two main elements. The first element is the inversion of single spectrum data for t^* data using the method of Pesicek et al. (2011) as described in Bisrat et al. (2014) and the inversion of spectral ratio data for differential t^* (dt^*) data using the method of Guo and Thurber (2021). The t^* and dt^* data are analogous to absolute and differential time data but contain path attenuation information. Both the single spectrum and spectral ratio inversion methods jointly invert for source and attenuation parameters. Because the spectral ratios can completely remove the instrument and site responses, the dt^* data from spectral ratio inversions are expected to be more accurate than the t^* data. The second element is tomographic inversion of the t^* and dt^* data for attenuation structure, which is analogous to DDV tomographic inversion of absolute and differential time data for velocity structure, but with earthquake locations fixed. Compared to the conventional Q tomography method using t^* data (e.g., Bisrat et al., 2014; Eberhart-Phillips & Chadwick, 2002; Pesicek et al., 2011; Thurber & Eberhart-Phillips, 1999), DDQ tomography can determine higher-resolution source-region attenuation structure due to the higher quality of dt^* data and the ability of dt^* data to cancel common path anomalies outside the source region (Guo & Thurber, 2021). We note that for the standard Q tomography method, as modified by Guo and Thurber (2021) to invert both t^* and dt^* data, we cannot separate intrinsic and scattering attenuation, and the resulting Q model represents a combination of intrinsic and scattering attenuation. See Texts S2, S3, and S4 in Supporting Information S1 for details on the methodology of the DDQ tomography and the spectral ratio inversion.

Both the DDV and DDQ tomographic inversions in this study were performed using the new tomoTD code (Guo & Thurber, 2021; Guo et al., 2021), which is a modified version of the tomoDD code (Zhang & Thurber, 2003) with some new features. tomoTD can use any combination of absolute data and event-pair, station-pair, and

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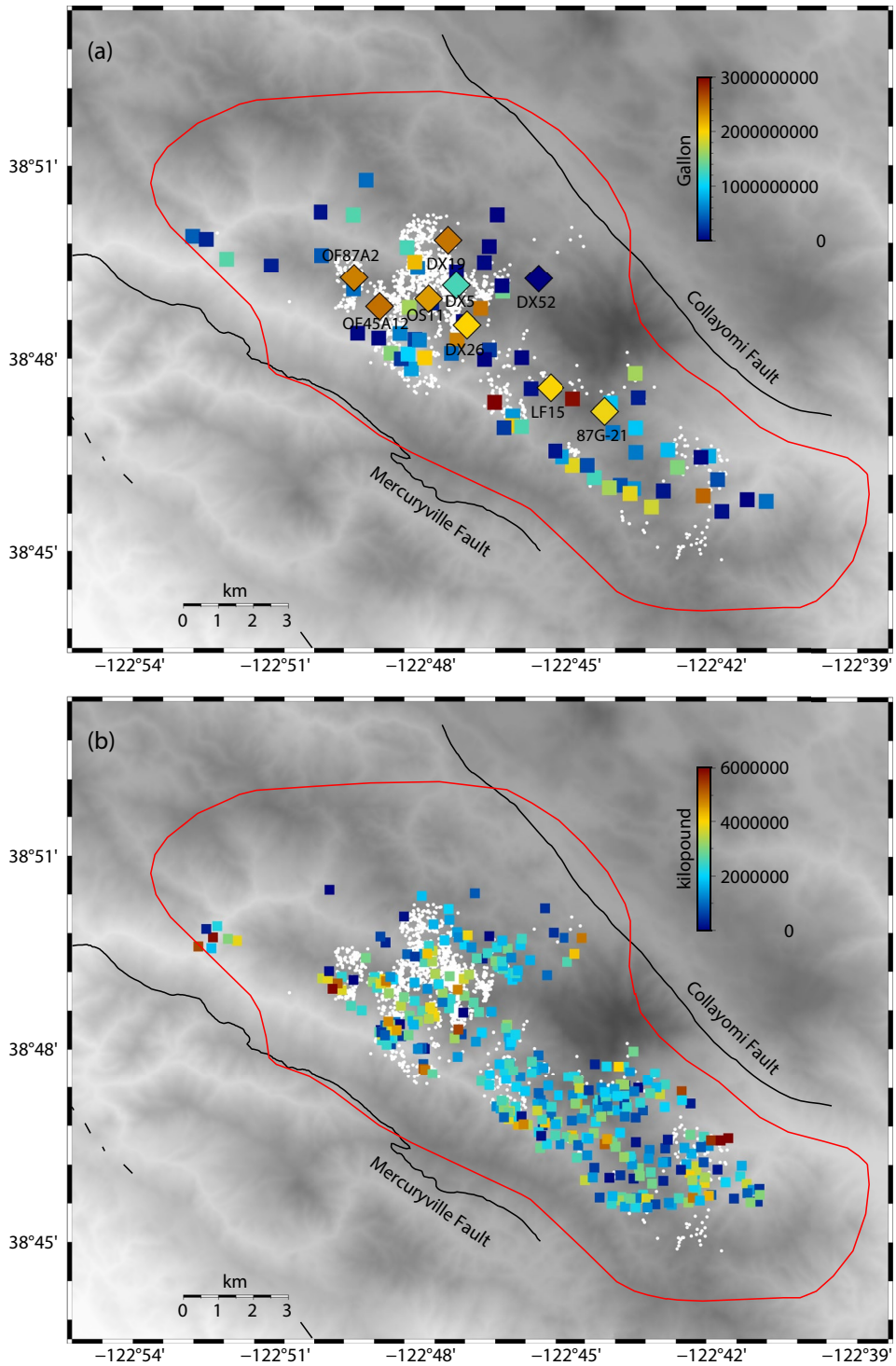


Figure 2. Map view of (a) water injection wells (squares and diamonds) and (b) steam production wells (squares). Only the wells with nonzero water injection and steam production between 2005 and 2011 are shown. Diamonds in (a) represent the water injection wells of the Santa Rosa-Geysers Recharge Project (SRGRP) with the well names labeled above or below them. Wells in (a) and (b) are colored by the total water injection volume and total steam production volume between 01 January 2005 and 31 December 2010, respectively. White dots represent the earthquakes in 2011 that are used in this study.

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double-pair differential data for locating seismic events (Guo & Zhang, 2017; Guo et al., 2017) and imaging velocity or attenuation structure (Guo & Thurber, 2021; Guo et al., 2021). For DDV and DDQ tomography, absolute data and event-pair differential data can be used as the input data. However, as shown in Guo and Thurber (2021), DDQ tomography with spectral ratio dt^* data alone is adequate to determine a high-resolution attenuation model of The Geysers due to the small size of The Geysers model volume. Thus, we used only the spectral ratio dt^* data in this study.

3. Tomographic Inversion Details

3.1. Selecting Closely Located Earthquakes and Common Recording Stations

One straightforward way to determine the temporal change of 3-D subsurface structure is to determine the individual models for each period separately and then calculate the model difference between different periods. In this study, we adopted this way to calculate the differences of the velocity and attenuation models between years 2005 and 2011. However, this approach can introduce model change artifacts if the individual models have inconsistent model sampling due to different earthquake locations and corresponding ray path distributions. To mitigate this problem, we carefully selected pairs of closely located earthquakes for the years 2005 and 2011 and common phases for the selected earthquake pairs, which yields similar model sampling for the two periods.

In a preliminary stage, we paired nearby earthquakes that occurred in 2005 and 2011 from the high-quality earthquake catalog of Gritto and Jarpe (2014) and selected ~2,900 earthquake pairs that have the most station observations in common. The selected earthquakes in 2011 have been used previously in Guo and Thurber (2021) for DDV tomography and DDQ tomography. As shown in Guo and Thurber (2021), however, the catalog earthquake locations from Gritto and Jarpe (2014) are somewhat scattered and can be significantly improved using DD location and velocity tomography methods (Waldhauser & Ellsworth, 2000; Zhang & Thurber, 2003) with improved arrival time data and WCC differential data. Since the accuracy of relative earthquake locations is critical for selecting closely located earthquakes, we relocated the selected earthquakes in 2005 using DD tomography. We used improved *P* wave arrival time data from an Akaike Information Criteria based arrival picking code (Guo et al., 2018) and more accurate WCC differential time data. With our earthquake relocations in 2005 and the earthquake relocations in 2011 from Guo and Thurber (2021), we finally selected 1,720 pairs of closely located earthquakes for the years 2005 and 2011 (Figure 1), requiring that each earthquake pair has an interevent distance <0.25 km. For each earthquake pair, we selected *P* wave arrival time data from common stations for DDV tomography and selected *P* wave t^* data from common stations for DDQ tomography.

3.2. DDV Tomographic Inversion

After the data selection described above, we obtained 17,795 absolute *P* wave arrival time data from 1,720 earthquake pairs observed at up to 23 stations (Figure 1). From the absolute arrival data, we constructed two sets of catalog differential time data, one using the ph2dt algorithm (Waldhauser, 2001) and the other one using a new algorithm based on a "best cell event" (bce) strategy (Guo & Thurber, 2021). These two algorithms are designed to produce sets of differential time data for determining high-precision relative earthquake locations and a high-resolution velocity model, respectively (Guo & Thurber, 2021). Guo and Thurber (2021) performed two separate inversions with these two sets of 2011 catalog differential data. The first inversion was to use DDV tomography with bce-derived differential data to determine a high-resolution velocity model. The second inversion was to use the DD location method with ph2dt-derived differential data and with the model from the first inversion to further constrain relative earthquake locations. In contrast to performing two separate inversions as in Guo and Thurber (2021), here we merged the two sets of catalog differential data for one inversion so that the high-resolution relative earthquake locations and velocity model can be better constrained at the same time. We obtained 801,100 catalog differential data for the 2005 earthquakes, consisting of 80,072 and 721,028 catalog differential data constructed with ph2dt and bce, respectively. We obtained 732,490 catalog differential data for the 2011 earthquakes, consisting of 79,699 and 652,791 catalog differential data constructed with ph2dt and bce, respectively.

In addition to the absolute and catalog differential time data, the 2005 and 2011 inversions also used 1,281,984 and 1,460,984 WCC differential time data, respectively, from pairs of earthquakes with hypocenters separated by

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<2 km observed at common stations. All the available stations (Figure S1 in Supporting Information S1) were used to measure WCC data for both years. The 2011 WCC differential data are from Guo and Thurber (2021). We followed the same approach as Guo and Thurber (2021) to determine the 2005 WCC differential data. Each earthquake pair was required to have at least eight differential time measurements with WCC coefficient larger than 0.7, and only those with WCC coefficient above 0.7 were used for tomographic inversions.

Figure 1 shows the Cartesian coordinate system and the horizontal grid nodes used for DDV tomography. The coordinate system is centered at (latitude 38.808°, longitude −122.784°). The grid spacing in both the longitude (X) and latitude (Y) directions is set at 0.6 km within the geothermal field and is larger near the boundaries of the field. The vertical nodes are positioned at −2, −0.8, 0, 0.4, 0.8, 1.2, 1.6, 2, 2.4, 2.8, 3.2, 3.6, 4, 4.4, 4.8, 5.2, 5.6, and 7-km depth relative to sea level. Both the 2005 and 2011 tomographic inversions used the Vp model determined by Guo and Thurber (2021) as the starting model. As in Zhang and Thurber (2003), a hierarchical weighting strategy was used to weight absolute arrival data and catalog and WCC differential data in successive stages of the inversion. The regularization parameters (damping and smoothing) were selected through a trade-off curve analysis (Aster et al., 2018; Guo & Thurber, 2021) to balance model roughness and data residual reduction. The damping values for different iterations were also adjusted to maintain the desired condition number of the inversion system, which indicates the stability of the inversion (Aster et al., 2018). Note that the inversion strategies and regularization parameters for the 2005 and 2011 inversions were exactly the same.

For the 2005 inversion, the RMS residual was reduced from 0.088 to 0.038 s for catalog data and from 0.089 to 0.008 s for WCC data. For the 2011 inversion, the RMS residual was reduced from 0.073 to 0.030 s for catalog data and from 0.090 to 0.014 s for WCC data.

3.3. DDQ Tomography Inversion

Guo and Thurber (2021) have described the details of the processing for the 2011 attenuation data, including the single spectrum inversion for t^* data, the spectral ratio inversion for dt^* data, and the quality control on the t^* and dt^* data. We processed the 2005 t^* and dt^* data in the same way that Guo and Thurber (2021) used for the 2011 data. We calculated the signal spectra from 1.024 s P wave time window of the unfiltered vertical component waveforms (0.424 s before and 0.6 s after the arrivals), which were then used to calculate spectral ratios. The noise spectra were calculated from 1.024 s time windows before the signal windows. Spectral ratios within a frequency range of 5.86–50 Hz were used for inversions. For the spectral ratio data of each station, we required the signal-noise-ratios of both spectra to be above 3 within a common and continuous frequency band with at least 10 Hz width. More details are described in Text S4 in Supporting Information S1. Figures S2 and S3 in Supporting Information S1 show the spectral ratio fitting results for two example event pairs. We then selected t^* data from common stations for the selected 2005 and 2011 earthquake pairs to ensure the same data distribution for both years. We obtained 13,105 t^* data for both years. We then selected dt^* data by requiring that both earthquake-station pairs for each dt^* datum must exist in the selected t^* data. We finally obtained 1,940,825 spectral ratio dt^* data from 1,677 earthquakes in 2005 and 1,868,714 spectral ratio dt^* data from 1,680 earthquakes in 2011, recorded by 23 common stations for both years. Some earthquakes and stations were excluded due to the quality control on the attenuation data and the common station requirement.

As described in Texts S3 and S5 in Supporting Information S1, our spectral ratio inversion method jointly inverts for dt^* and earthquake corner frequency (f_c) and can provide f_c estimates (Figure S4 in Supporting Information S1). One major limitation for reliably estimating f_c is the frequency bandwidth (Abercrombie et al., 2017; Ruhl et al., 2017). Figure S5 in Supporting Information S1 shows that the upper limits of useable frequency ranges of spectral ratios for most event pairs we studied are at least two times higher than the inverted f_c, even for small magnitude earthquakes (Figure S6 in Supporting Information S1). But taking into account the uncertainty/controversy over the reliability of f_c estimates and the associated stress drop scaling (see review by Abercrombie (2021)), we think more work in the future is required to validate the reliability of our f_c estimates. We note that the dt^* measurements from our spectral ratio inversions, which are the focus of this study, are not significantly affected by the recovery of f_c based on the synthetic tests of Guo and Thurber (2021).

For the 2005 and 2011 DDQ tomographic inversions, the coordinate system and grid set-up are the same as that used for DDV tomography. The 2005 and 2011 inversions started from the same 1-D gradient Qp model as used

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by Guo and Thurber (2021) (Table S1 in Supporting Information S1). We also performed inversions starting from the 3-D Qp model of Guo and Thurber (2021) and obtained very similar results, indicating that DDQ tomography is not very sensitive to the starting model. The optimal damping and smoothing values were estimated through a trade-off curve analysis (Aster et al., 2018; Guo & Thurber, 2021) to balance model roughness and data residual reduction. The damping values for different iterations were also adjusted based on the condition number of the inversion system. After the 2005 and 2011 inversions, the RMS residual was reduced from 0.0089 to 0.0075 and from 0.0083 to 0.0069, respectively. After testing numerous inversions using a wide range of regularization parameter values, we believe our final misfit has reached the level of data errors, although the residual reduction is somewhat modest.

4. Tomographic Results, Resolution Tests, and Uncertainty Analysis

Figures 3 and 4 show cross-sections of the final Vp and Qp models for 2005 and 2011 along profiles AB-GH shown in Figure 1, crossing most SRGRP injection wells in the southeastern part of the northwest Geysers. Figure 5 shows the model differences between the 2 years (dVp and dQp) by subtracting the 2005 models from the 2011 models.

To assess the model resolution, we performed checkerboard tests with and without noise. For the checkerboard test of the Vp model, we first constructed the true Vp checkerboard models by adding positive and negative 3% anomalies applied to nodes of the starting model in a 3-by-3-by-3 pattern (Figures S7 and S8 in Supporting Information S1). The patterns of positive and negative anomalies in 2005 and 2011 models were opposite, which gives the true dVp model a checkerboard pattern with 6% anomalies (Figure S9 in Supporting Information S1). With the true checkerboard model, we generated synthetic data with the same data distribution as the real data. For the noise-added test, we added Gaussian random noise with zero mean and a standard deviation of 0.02 s for catalog arrival time data and 0.01 s for WCC differential data. Finally, we inverted the synthetic data with the same starting models and inversion parameters as the real data inversions. Figures S7 and S8 in Supporting Information S1 show the recovered 2005 and 2011 Vp models, respectively. The dVp models show that the reservoir at depths of ~0.5–4 km is recovered well in the areas of good data coverage (Figure S9 in Supporting Information S1). To test if anomalies of smaller size can be resolved, we further performed a set of checkerboard tests with the true checkerboard models constructed in a 2-by-2-by-2 pattern (Figures S10–S12 in Supporting Information S1). In general, the well recovered regions estimated from the 2-by-2-by-2 checkerboard tests are similar to that from the 3-by-3-by-3 checkerboard tests.

For the checkerboard test of the Qp model, the processing is similar to the Vp model test but with several differences. (a) The true positive and negative Qp anomalies have an amplitude of 30% and the resulting true dQp model anomalies have an amplitude of 60%. (b) The Gaussian noise has a standard deviation of 0.006 for the noise-added synthetic dt^* data. Figures S13–S15 in Supporting Information S1 show the recovered Qp and dQp models from the 3-by-3-by-3 tests and Figures S16–S18 in Supporting Information S1 show the results from the 2-by-2-by-2 tests. In general, the regions with well recovered Qp and dQp are similar to the regions with well recovered Vp and dVp.

To quantitatively estimate model resolution, we further calculated the model resolvability with the checkerboard test results based on a method modified from that of Zelt (1998) and described in Guo et al. (2021). For each node, its resolvability is defined as the semblance between the true and recovered checkerboard model values of that node and its surrounding nodes. Figures S7–S18 in Supporting Information S1 show the contours of model resolvability of 0.7 for the recovered checkerboard models, which approximately outline the well recovered regions. In the real data inversion model shown in Figures 3–6, the model resolvability values estimated from the noise-free checkerboard tests are used to mask the poorly resolved regions.

We estimated the uncertainties of earthquake locations and tomographic models using the bootstrapping method, a statistical resampling method (Efron & Gong, 1983; Efron & Tibshirani, 1991; Guo & Zhang, 2017; Guo et al., 2021). For each of the 2005 and 2011 Vp and Qp tomographic inversions, we randomly resampled its input data and then performed the inversion using the same parameters as that used in the original inversion. We repeated this process 50 times to obtain good estimates of standard deviations for each event location and the model value for each node, which were used as the estimates of uncertainties. Our bootstrapping analysis shows that the median location uncertainties are 15, 26, and 36 m for 2005 earthquakes in latitude, longitude, and depth,

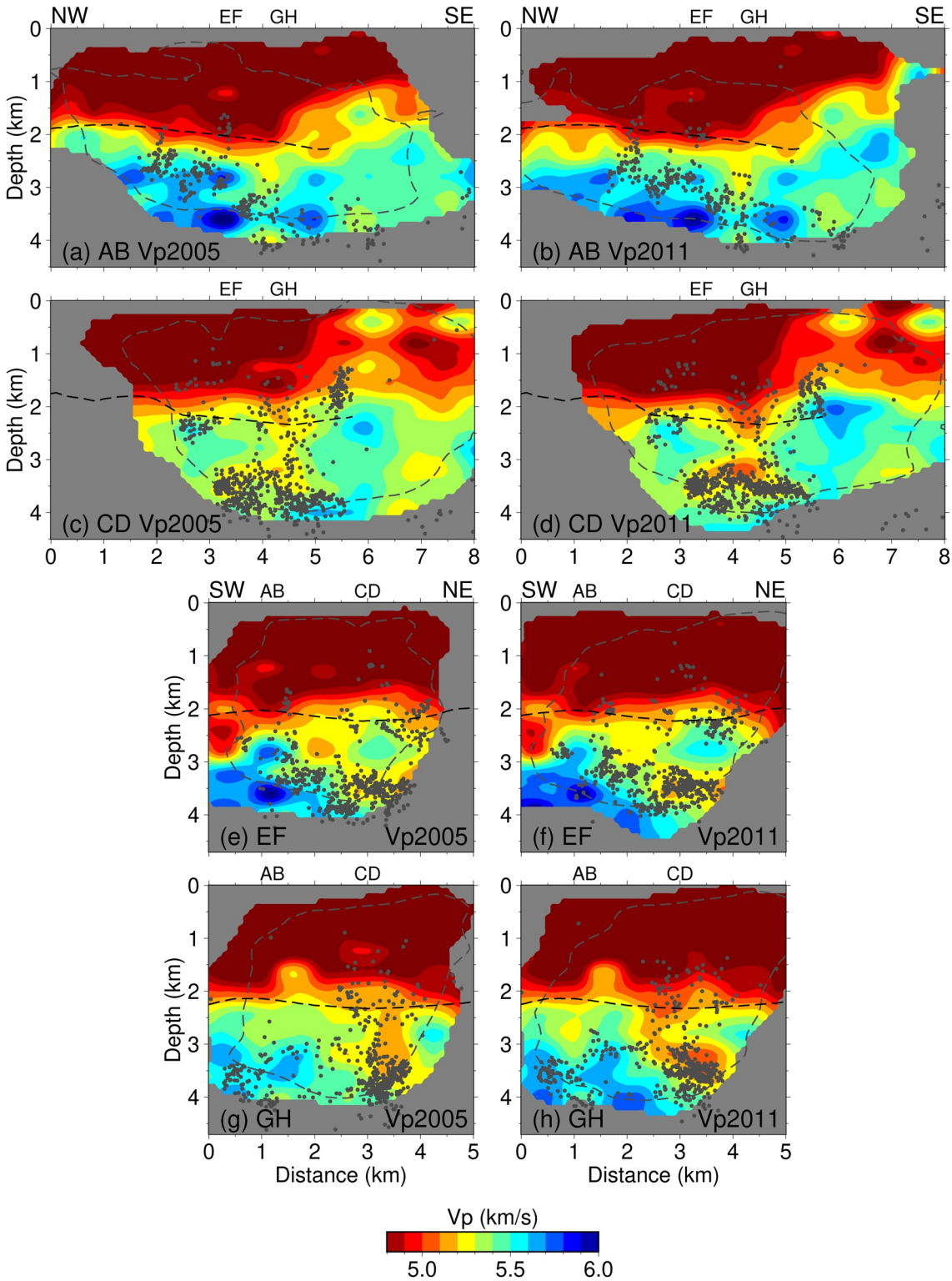


Figure 3. Cross-sections of Vp models and earthquakes (dots) of (a, c, e, g) year 2005 and (b, d, f, h) year 2011 along profiles (a–b) AB, (c–d) CD, (e–f) EF, and (g–h) GH shown in Figure 1. Earthquakes within 0.6 km of each cross-section are shown. Black dashed lines at ~2-km depth represent the top of the high temperature reservoir (HTR). Poorly resolved regions, estimated from the 3-by-3-by-3 checkerboard tests, are masked. Gray dashed lines outline the higher-resolution regions estimated from the 2-by-2-by-2 checkerboard tests. Where the other profiles cross each cross-section are noted across the top of each panel.

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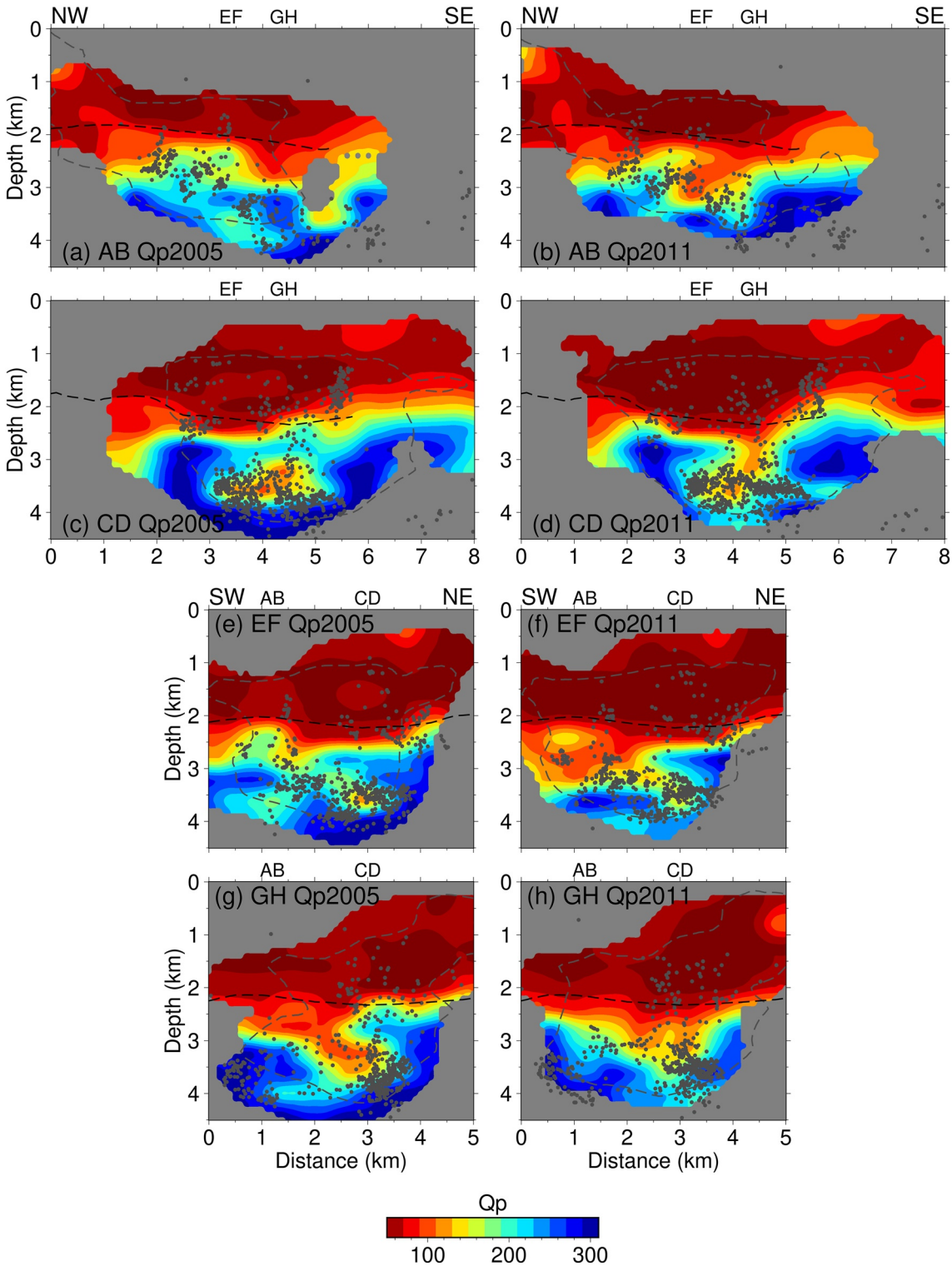


Figure 4. Cross-sections of Qp models and earthquakes (dots) of (a, c, e, g) year 2005 and (b, d, f, h) year 2011 along profiles (a–b) AB, (c–d) CD, (e–f) EF, and (g–h) GH shown in Figure 1. Earthquakes within 0.6 km of each cross-section are shown. Black dashed lines at ~2-km depth represent the top of the high temperature reservoir (HTR). Poorly resolved regions, estimated from the 3-by-3-by-3 checkerboard tests, are masked. Gray dashed lines outline the higher-resolution regions estimated from the 2-by-2-by-2 checkerboard tests. Where the other profiles cross each cross-section are noted across the top of each panel.

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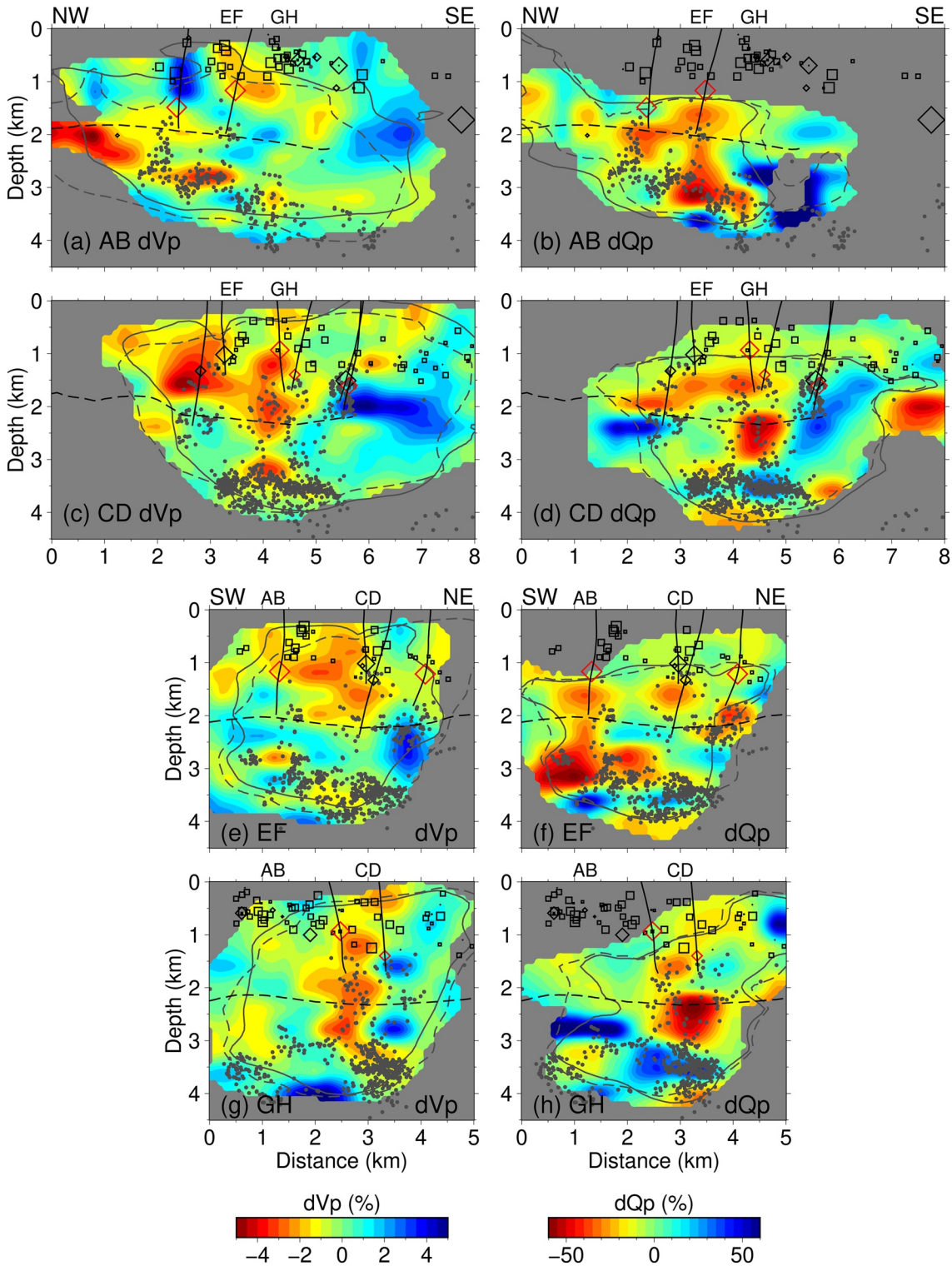


Figure 5.

respectively, and 13, 20, and 27 m for 2011 earthquakes. For the regions of high resolution estimated by checkerboard tests, the median uncertainties are 0.03 and 0.02 km/s for the 2005 and 2011 Vp models, respectively, and are 0.9 and 0.9 for the 2005 and 2011 Qp models, respectively. Cross-sections of the estimated uncertainties of the Vp and Qp models are shown in Figures S19 and S20 in Supporting Information S1 along the same profiles as

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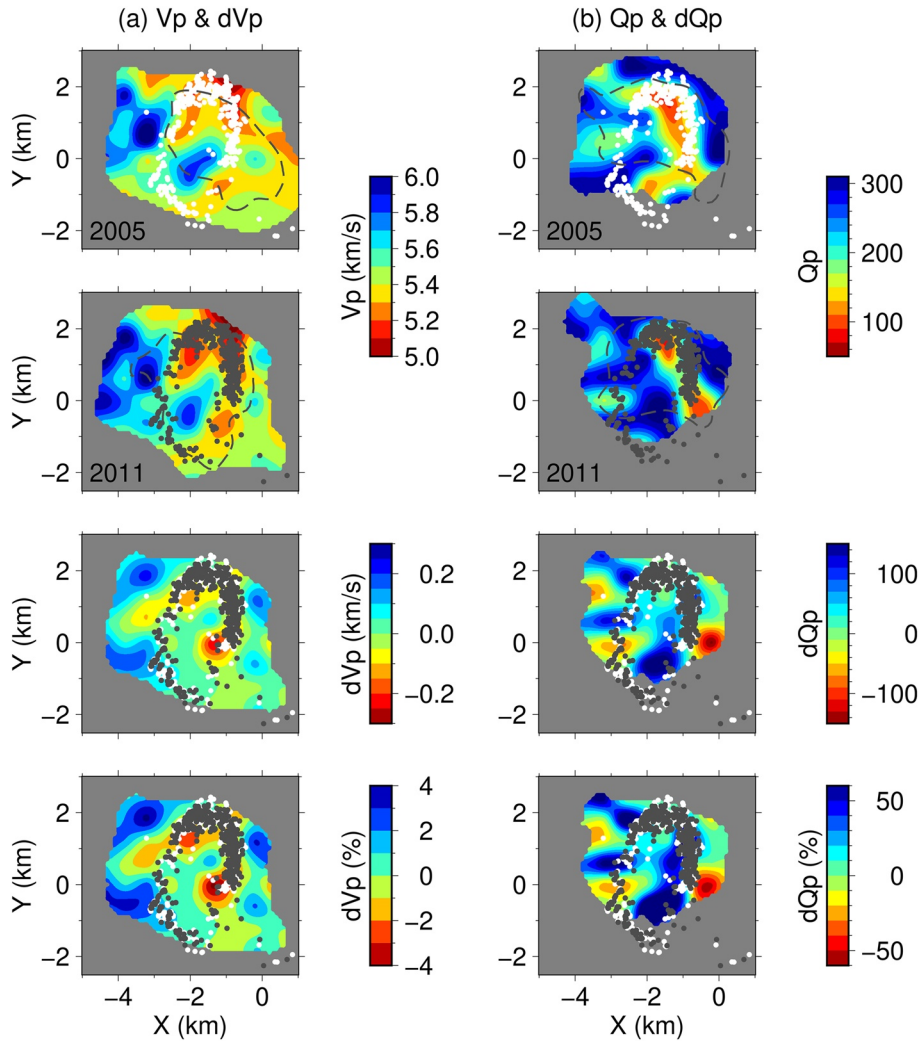


Figure 6. Horizontal slices of Vp and Qp models and model changes (dVp and dQp) at 3.6-km depth. (a) Vp and dVp models. (b) Qp and dQp models. dVp and dQp models in the third and fourth rows are shown in two different ways. The first is the absolute change between 2011 and 2005 models. The second is the percent change of the 2011 models from the 2005 models. Poorly resolved regions, estimated from the 3-by-3-by-3 checkerboard tests, are masked. In Vp and Qp model slices, dashed lines outline the higher-resolution regions estimated from the 2-by-2-by-2 checkerboard tests. White and gray dots represent 2005 and 2011 earthquakes within 0.2 km of the depth slice, respectively.

that shown in Figures 3 and 4. We estimated the uncertainties of the dVp and dQp models by dividing the square root of the sum of the square of the 2005 and 2011 model uncertainties by the 2005 model values, the results of which are shown in Figure S21 in Supporting Information S1. The maximum values of the estimated dVp and dQp uncertainties for the regions of high resolution are ~1% and ~5%, respectively (Figure S21 in Supporting

Figure 5. Cross-sections of Vp and Qp model changes (dVp and dQp) along profiles (a–b) AB, (c–d) CD, (e–f) EF, and (g–h) GH shown in Figure 1. (a, c, e, g) dVp and (b, d, f, h) dQp are the percent changes of the 2011 models from the 2005 models. Only the well resolved regions in both 2005 and 2011 models, estimated from the 3-by-3-by-3 checkerboard tests, are shown. Solid and dashed gray lines outline the higher-resolution regions of the 2005 and 2011 models, respectively, estimated from the 2-by-2-by-2 checkerboard tests. Black dashed lines at ~2-km depth represent the top of the high temperature reservoir (HTR). Diamonds represent the water injection centers for the injection wells within 0.6 km of the cross-sections, among which the red ones represent the injection wells of the Santa Rosa-Geysers Recharge Project (SRGRP). The trajectories (solid lines) are shown for SRGRP wells and several other injection wells. Squares represent the steam production centers for the steam production wells within 0.6 km of the cross-sections. Diamonds and squares are scaled by the water injection volume and steam production volume between 01 January 2005 and 31 December 2020, respectively. The injection and production centers are determined by taking a weighted average of the steam entry measured depths and their pressure increases at each of those depths. Gray dots represent 2011 earthquakes within 0.6 km of each cross-section. Note that 2005 earthquakes shown in Figures 3 and 4 are not shown in this figure. Where the other profiles cross each cross-section are noted across the top of each panel.

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Information S1), which are substantially smaller than the amplitudes of the dVp and dQp anomalous bodies (Figure 5) that we will interpret.

Our Vp and Qp models for both years show strong lateral and vertical variations. Our models show sharp gradients at ~2-km depth with lateral variations, which well characterize the rapid temperature transition from the NTR to the HTR at depths of ~1.8–2.4 km (Figures 3 and 4; Walters et al., 1988). The lateral contrasts in both Vp and Qp in the NTR at a distance of ~4.5–5.5 km shown in cross-sections AB and CD may indicate a discontinuity in hydrologic properties separating the northwest Geysers reservoir and the reservoir to the south (Figures 3 and 4). This discontinuity may correspond to a northeast-trending fault or shear zone (Guo & Thurber, 2021), which is labeled the Cobb Creek Fault Zone in Figure 1 of Garcia et al. (2016). In the HTR, the earthquake generation zones show strong low Vp and low Qp anomalies for both years.

Our dVp and dQp models shows substantial depth variations below the injection centers, associated with variations in the degree of clustering of 2011 microearthquakes (Figure 5). At <~3-km depth below most injection centers covering the NTR and the upper part of the HTR, both Vp and Qp decreased from 2005 to 2011 (Figure 5). Figures 5c and 5d show that the 2011 earthquakes are broadly distributed at distances of ~3.5–5 km and depths of ~1.5–3 km associated with substantially decreased Vp and Qp, which also can be seen in Figures 5g and 5h at a distance of ~3 km. The lower part of the HTR at a depth of ~3.5 km below the injection centers shows anomalous zones with decreased Vp, increased Qp (Figures 5 and 6), and more tightly clustered or linearly distributed earthquakes compared to the broadly distributed earthquakes in the overlying region (Figures 5c-5d, 5g–5h, and 6). The exceptions are the region below DX26 at a depth of ~2 km and at a distance of ~5.5 km along CD, where both Vp and Qp increased (Figures 5c and 5d), and the region below OF45A12 at depths of ~1.2–3.5 km and at distances of ~0.5–2 km along EF, where Vp is unchanged and Qp decreased (Figures 5e and 5f).

Considering that the seismicity patterns from our small catalog selected for tomography may not be representative, we also show earthquake locations from the much more complete, high-precision DD catalog from the NCEDC (NCEDC, 2014; Waldhauser & Schaff, 2008), on the dVp and dQp model cross-sections (Figures 7–9). Overall, the absolute earthquake locations from the complete DD catalog (Figures 7–9) are similar to our small catalog (Figures 1 and 5), which allows us to use this DD catalog for the discussions of seismicity and the relationship between seismicity and the velocity and attenuation models. The seismicity patterns seen from our small catalog, including the broad distribution of 2011 seismicity in the NTR and the upper part of the HTR and the concentration of 2011 seismicity in the lower part of the HTR, are also present in the complete DD catalog (Distance = ~3–5 km in Figures 8g and 8h and Distance = ~2–4 km in Figures 9g and 9h). However, the complete DD catalog shows more features that cannot be seen from our small catalog. Some earthquake clusters in 2011 beneath some injection wells, which are not present or present diffusely in our small catalog, can be seen clearly, e.g., the ones at ~2.5-km depth and at distances of ~0.5 and ~2.5 km along CD (Figures 8g and 8h) and the one at ~2.5-km depth and at a distance of ~4 km along EF (Figures 9c and 9d). The complete DD catalog also shows that some 2005 earthquake clusters beneath some injection wells, including the one at depths of ~2–3 km and at a distance of ~1.3 km along EF (Figures 9a and 9b) and the one at depths of ~1.5–3 km and at a distance of ~2.6 km along GH (Figures 9e and 9f), disappeared or became broadly distributed in 2011 (Figures 9c–9d and 9g–9h).

5. Discussion

In this section, we discuss how our tomographic dVp and dQp models have been used to understand fracturing and saturation states and their temporal changes in the NTR and HTR in the southeastern part of the northwest Geysers.

Mechanisms responsible for velocity and attenuation changes in the crust have been studied both theoretically and experimentally (Hutchings et al., 2019; Kuster & Toksöz, 1974; O'Connell & Budiansky, 1977; Toksoz et al., 1976; Winkler & Nur, 1979). Injection-induced processes responsible for Vp and Qp changes in geothermal reservoirs mainly include (a) inclusion of fluids with partial or full saturation, (b) increased rock density from fluid saturation, (c) increased rock damage due to rock thermal contraction, and (d) increase in pore pressure and decrease in effective stress. Figure S22 in Supporting Information S1 shows a schematic diagram that briefly depicts these processes and their effects on Vp and Qp. In high porosity rocks, the inclusion of fluids with partial saturation can increase the rock density so that Vp is decreased although the bulk modulus is increased

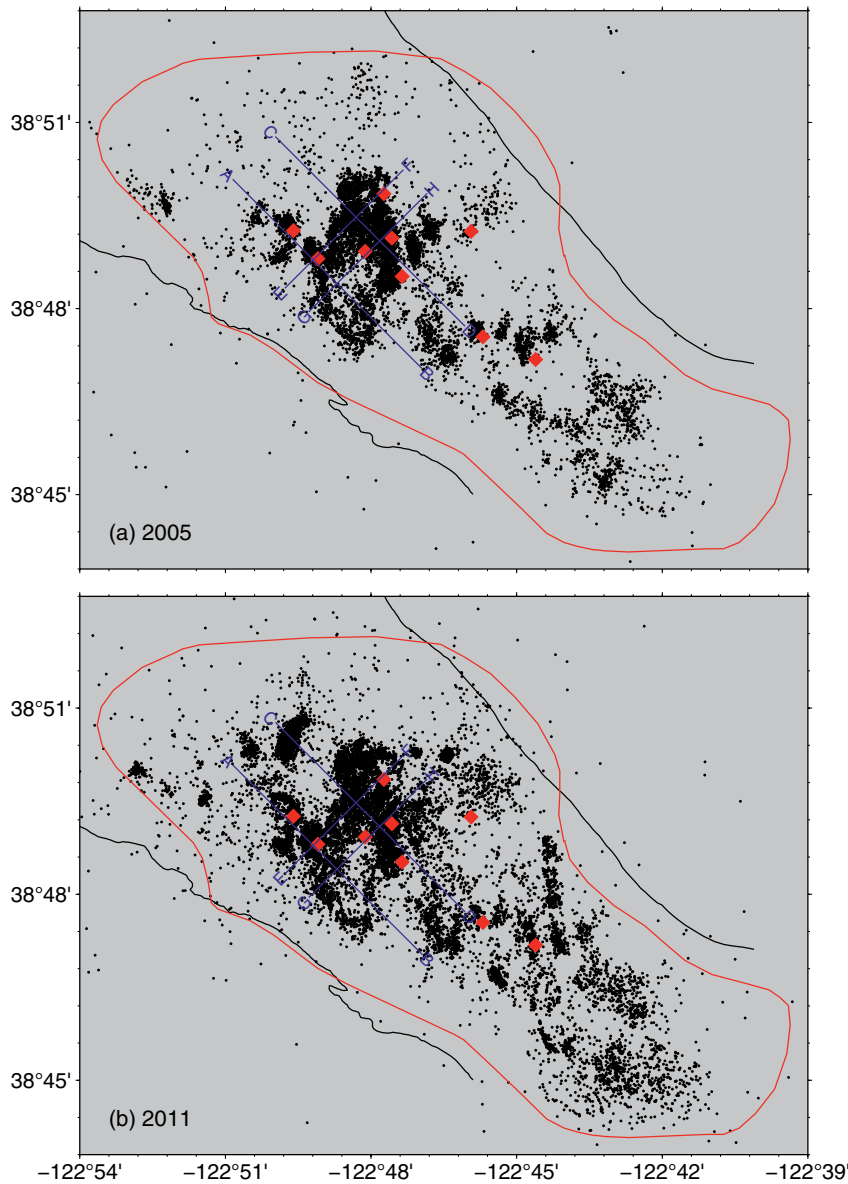


Figure 7. Map views of earthquake locations of (a) year 2005 and (b) year 2011 from the double-difference (DD) catalog from the Northern California Earthquake Data Center (NCEDC). Compared to the small catalog selected for tomography in this study (Figure 1), the DD catalog from the NCEDC is much more complete. Diamonds represent the injection wells of the Santa Rosa-Geysers Recharge Project (SRGRP).

(Hutchings et al., 2019). Increased but still partial fluid saturation can increase attenuation and thus decrease Qp. Increasing saturation to a nearly full state would decrease *P* wave attenuation due to the suppression of fluid flow between pores and thus increase Qp (Winkler & Nur, 1979). Vp is also increased when reaching full saturation (Winkler & Nur, 1979). However, the change from partial to full saturation has a much larger influence on Qp than Vp (Winkler & Nur, 1979). Injecting cold water into hot reservoir can cause rock contraction and thus more fracturing. More fracturing can decrease Vp and Qp. Earthquake ruptures may create new cracks resulting in higher permeability, which would decrease Vp and Qp. Increased fluid pressure and decreased effective stress can decrease both Vp and Qp (Winkler & Nur, 1979).

Our results show that the earthquake generation zones below the injection wells in the NTR and HTR are generally associated with prominent dVp and/or dQp changes with lateral and vertical variations (Figure 5). In the HTZ, the dVp and dQp anomalies correspond to low Vp and low Qp zones in both the 2005 and 2011 models

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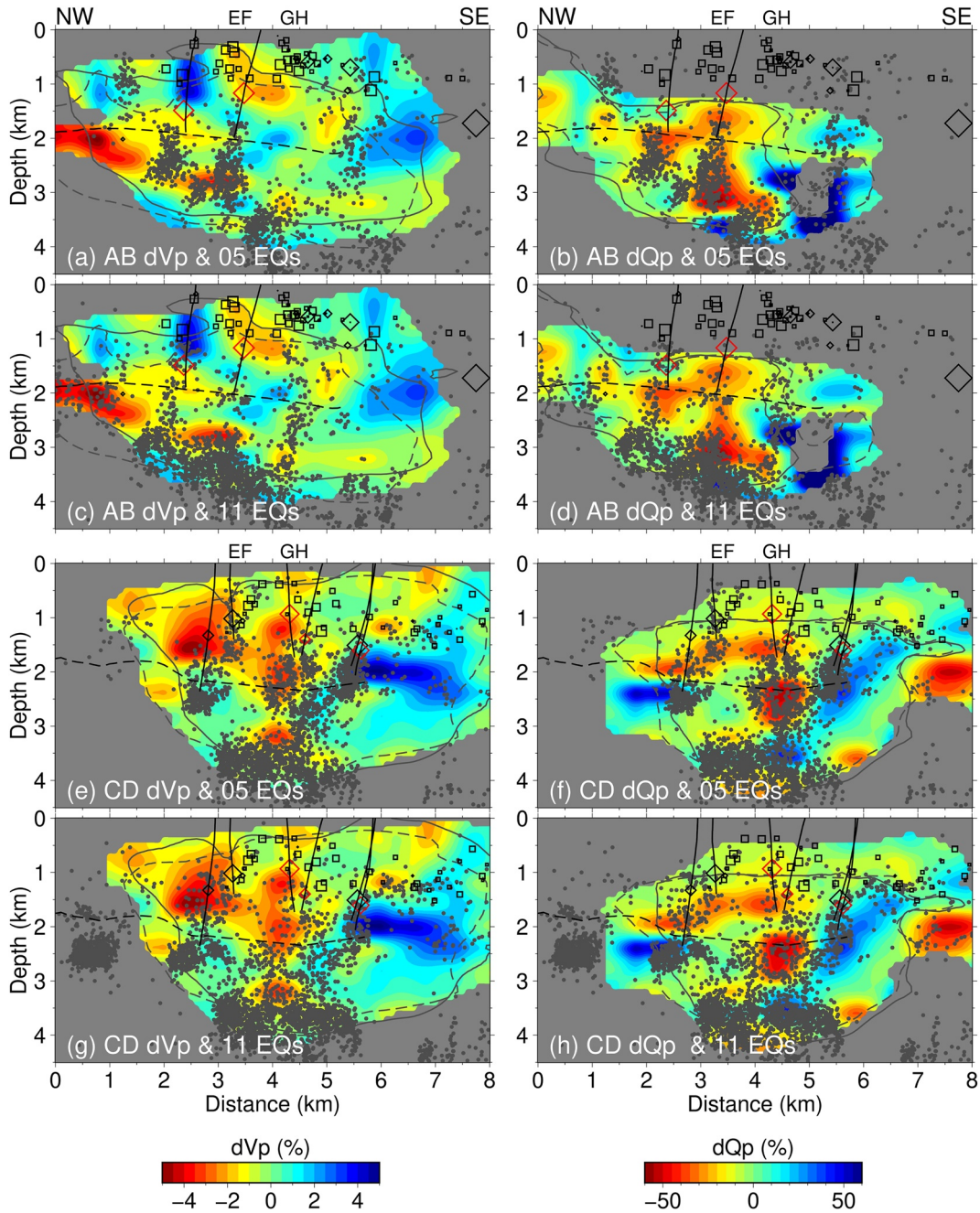


Figure 8. Cross-sections of earthquake locations and percentage changes of (left panels) Vp and (right panels) Qp models between 2005 and 2011. Compared to Figures 3–6, the earthquake locations shown in this figure are from the more complete double-difference (DD) catalog from the Northern California Earthquake Data Center (NCEDC). The panels (a), (b), (e), (f), (i), (j), (m), and (n) show 2005 earthquake locations. The panels (c), (d), (g), (h), (k), (l), (o), and (p) show 2011 earthquake locations. Only earthquakes with magnitude above 0.4 and within 0.6 km of each cross-section are shown. The other elements are the same as Figure 5.

(Figures 3–5). These observations suggest that the earthquakes and dVp and dQp changes below ∼1–1.5 km are mainly induced by water injection with additional contributions from steam production in the NTR. This is supported by many previous studies of The Geysers. Stark (2003) found strong spatial and temporal correlations between water injection and deep earthquake clusters in the HTZ. Gritto and Jarpe (2014) found positive correlations between variations in water injection volume and seismic Vp/Vs in the reservoir. Seismic tomography and mechanical modeling studies found that water injection has a significant effect on the physical and mechanical

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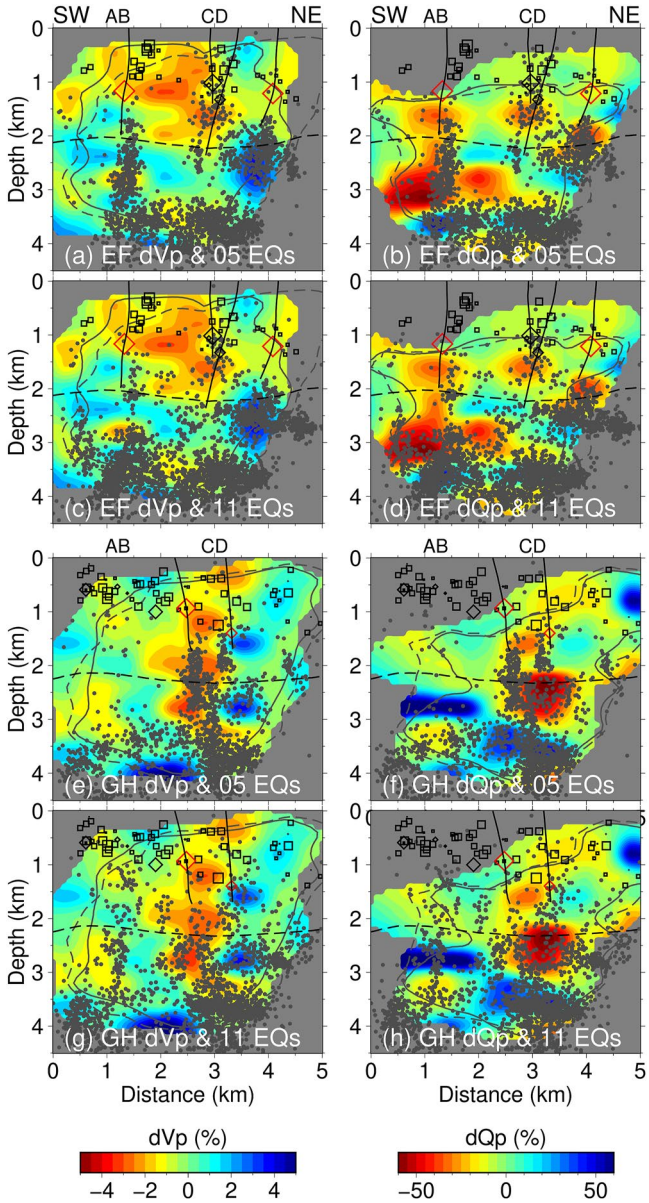


Figure 9. Same as Figure 8 but for another two cross-sections (EF and GH).

properties of the reservoir rocks in the region surrounding the wellhead and extending downward (Hutchings et al., 2014; Jeanne et al., 2015; Rutqvist & Oldenburg, 2008; Rutqvist et al., 2016). Stark (2003) pointed out that the deep seismicity in the HTZ within our study area has been observed since 1976. The long injection history accounts for the strong low Vp and low Qp anomalies in the HTR associated with active seismicity (Figures 3 and 4). Guilhem et al. (2014) performed moment tensor inversions for the $M \sim 3$ earthquakes located in both the NTR and HTR within our study area and found that most of the $M \sim 3$ earthquakes have composite mechanisms with significant nondouble-couple components, although the double-couple components are still dominant. The mixed shear and tensile faulting indicate rapid fluid flow accompanying crack opening in the reservoir due to water injection (Guilhem et al., 2014; Ross et al., 1996). Detailed studies of a seismicity cluster occurring in the HTR surrounding and below the wellheads of two injection wells to the north of our study area found clear long-term and short-term spatiotemporal correlations between many source characteristics of the induced seismicity and the associated local stress state with water injection rates (Kwiatek et al., 2015; Martínez-Garzón et al., 2013, 2014; Staszek et al., 2017). Kwiatek et al. (2015) found no visible relation between this induced seismicity cluster with steam production. Moment tensor inversions for the earthquakes in this cluster also found significant nondouble-couple components that were suggested to be related to water injection, local stress state, and faulting regime (Martínez-Garzón et al., 2017).

How injection and production change the reservoir conditions in the northwest Geysers has been investigated by some thermal-hydrological-mechanical modeling studies. Rutqvist and Oldenburg (2008) modeled the long-term (44 years) effects from a pair of injection and production wells extending into the top of the HTR and found significant reservoir condition changes. Their modeling results show that injection causes the formation of a wet and cooling zone near the wells, extending from the NTR downwards to the HTR. The wet, cooling zone is elongated vertically, similar to some of our observed dVp and/or dQp change zones below the injection wells (Figure 5). The production-induced pressure depletion is mitigated partly in the NTR and completely in the HTR by injection. The model by Rutqvist and Oldenburg (2008) shows that within the cooling zone, the reservoir shrinkage caused by the cooling results in a reduction of vertical effective stress, which drives the cooling zone into failure. Rutqvist and Oldenburg (2008) also modeled the 6-month effects of injection and production with seasonal variations and found significant changes 1 km below the wellhead in the HTR, including increased fluid content, increased fluid pressure, decreased temperature, and decreased effective stress. Jeanne et al. (2015) modeled effects over 1–2 months from an injection well during an EGS Demonstration Project in the northwest Geysers and found similar changes in the region surrounding the injection part of the well, which is centered near the NTR-HTR transition depth.

Injection-induced rapid fluid flow accompanying crack opening and the activation of critically stressed existing fractures due to stress changes induced by rock thermal contraction by injecting cool waters into hot rocks is likely to be the dominant mechanism inducing microearthquakes at The Geysers, with contributions from increased fluid pressure and poroelastic stress changes activating critically stressed existing fractures (Guilhem et al., 2014; Martínez-Garzón et al., 2014; Ross et al., 1996; Rutqvist & Oldenburg, 2008). Martínez-Garzón et al. (2014) estimated the diffusivities of thermal and hydraulic pressure perturbations and the corresponding characteristic diffusion lengths from one injection wellhead in the northwest Geysers and found that the thermal fracturing could induce a strong stress change for inducing earthquakes near the wellhead, whereas the pore pressure diffusion might be responsible for small stress changes inducing earthquakes at greater distances from the wellhead.

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More broadly distributed 2011 seismicity in the NTR and the upper part of the HTR probably indicates more diffuse fluid flow through the prevalent fracture network with pre-existing and/or newly created small-scale fractures in 2011 (Figures 5c–5d, 5g–5h, 8g–8h, and 9g–9h). The two vertically distributed earthquake clusters in 2005 in the NTR and upper part of the HTR beneath two SRGRP injection wells (the one at a distance of ∼1.3 km along EF in Figures 9a and 9b and the one at a distance of ∼2.5 km along GH in Figures 9e and 9f) disappeared or became broadly distributed in 2011 (Figures 9c–9d and 9h). This suggests that the fracture zones and fluid paths in the NTR and upper part of the HTR beneath the two wells were narrow in 2005 but broadened in 2011 after 6 more years of injection. The more concentrated and linearly distributed seismicity in the lower part of the HTR in 2011 (5c–5h, 6, 8d,8f and 8h) probably indicates more localized, confined fluid flow there along fracture/fault zones that are likely pre-existing, although small-scale fractures could be created as fluids flow in the pre-existing fracture/fault zones. The orientations of faults/fractures in the HTR seen from the relocated seismicity in our resolved area (Figure 6) are consistent with that found in other parts of the northwest Geysers by previous studies. Northwest-striking and northeast-striking pre-existing shear zones with a length of ∼1 km have been inferred in the HTR to the north of our study area based on induced seismicity analysis associated with the Northwest Geysers EGS demonstration project there (Jeanne et al., 2014). Critically stressed fractures preferentially parallel to the maximum horizontal stress direction (NNE-SSW) and a local fault striking in the NNW-SSE direction in the HTR below another two injection wells to the north of our study area have also been found by Martínez-Garzón et al. (2014). These were suggested to be responsible for generating induced earthquakes due to combined effects of thermal fracturing, increased pore pressure, and poroelastic stress changes based on the analysis of the spatiotemporal, kinematic, and source characteristics of the induced seismicity during different periods of injection. Analyzing fault geometries from the same seismicity cluster, Martínez-Garzón et al. (2016) found the majority of the activated fractures had orientations that were favorable for failure with respect to the stress field. Hydrofracturing is a common explanation for induced seismicity and fractures associated with the exploitation of tight oil/gas but is unlikely for The Geysers considering that all the injected waters at The Geysers are not pumped but flow into the reservoir from the wellheads at low pressure (Garcia et al., 2016).

Our observed changes of decreased Vp and Qp in the NTR at depths of ∼1–2.2 km and in the upper part of the HTR at depths of ∼2.2–3 km, associated with broadly distributed seismicity in 2011 (Figures 5c–5d, 5g–5h, 8g–8h, and 9g–9h), can be explained by the increase in rock density due to increased but still partial saturation, reduction in modulus due to increased fracturing, and reduction in effective stress, all of which could be caused by water injection and the resulting rock contraction. Production-induced pressure depletion in the NTR may also contribute to reservoir shrinkage. For the zones of concentrated and linearly distributed earthquakes in 2011 at ∼3.5-km depth in the HTR (Figures 5c–5h, 6, 8g–8h, 9c–9d, and 9g–9h), the decrease in Vp can be explained by the increased fracturing and reduction in effective stress, whereas the increase in Qp may indicate a change from partial steam/water saturation to full fluid saturation in the fracture zones. We note that increased fracturing and decreased effective stress also decrease Qp and full saturation also increases Vp, so the relative contributions of these mechanisms may control our observed changes of decreased Vp and increased Qp in the lower part of the HTR. Thus, we suggest that the key reason for whether Qp decreases or increases may lie in the degree of fluid saturation.

Geology and core studies suggest that the northwest Geysers is composed of hydrothermally altered and fractured metagraywacke of high permeability in the NTR and low-porosity, low-permeability hornfelsic metagraywacke and granite in the HTR (Garcia et al., 2016; Lutz et al., 2012; Walters, 2013). These are consistent with our observations. High permeability can result in more diffuse fluid flow in the NTR whereas low permeability can result in more localized, confined fluid flow in the lower part of the HTR along some fracture/fault zones which are more permeable relative to the host rocks. Thus, continuous injection for multiple years can result in full saturation in the lower part of the HTR, whereas the NTR can still be partially saturated. Decreased Vp and increased Qp in the low-permeability HTR after a 2-month injection during the EGS Demonstration Project to the northwest of our resolved area were also observed by Hutchings et al. (2014) and Rutqvist et al. (2016). However, our observed decreased Vp and Qp and broadly distributed seismicity pattern in the upper part of the HTR indicate high permeability there. The difference in permeability between the upper part of the HTR in the southeast part of the northwest Geysers, i.e., our resolved area, and the HTR to the northwest of our resolved area may explain the success of the inadvertently created EGS and thus high steam productivity in our resolved area compared to the other parts of the northwest Geysers (Stark, 2003). The pre-existing permeable fault zones in the lower part of the HTR, which serve as fluid-flow pathways, may also contribute to the success of EGS in this area.

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In addition to the major changes interpreted above, there are two other changes below two SRGRP injection wells that may indicate different fracturing and saturation states. One area has increased Vp and Qp at 2-km depth and at a distance of ~5.5 km along CD in 5c–5d and 8e–8h. The other has decreased Qp and no significant change in Vp at depths of ~1.2–3.5 km and at distances of ~0.5–2 km along EF in Figures 5e–5f and 9a–9d. For the former, a possible interpretation is that a change from partial to full saturation dominates both Vp and Qp changes, which is supported by the clustered seismicity indicating a localized fracture zone there that can confine fluids. For the latter, the increased fracturing, increased partial saturation, and reduction in effective stress could explain the Qp change while their effects on Vp may be mitigated by the increase in bulk modulus due to the replacement of vapor with water that would increase Vp. Hutchings et al. (2019) suggest that saturated fractured rocks with low porosity may also explain decreased Qp and unchanged Vp. They note that Qp decreases because an increase in scattering attenuation contributes to the Qp change more than a decrease in intrinsic attenuation and Vp is unchanged due to the competing effects of decreased shear modulus and increased bulk modulus.

Our results show that the DDV and DDQ tomography methods can effectively determine high-resolution injection-induced medium changes of geothermal reservoirs from a long-term perspective (6 years here). However, there is also a high demand from the industry to monitor short-term reservoir medium changes to evaluate and adjust operation strategies. Our results can provide a rough estimate for the minimum period required to perform 4-D tomography to detect robust medium changes, which mainly depends on the relative amplitude of the medium changes and the tomographic model uncertainties. Our results show that the injection-induced Vp and Qp changes between 2005 and 2011 have amplitudes of ~3% and ~40%, respectively (Figure 5). By taking into account the estimated dVp and dQp model uncertainties of ~1% and ~5% (Figure S21 in Supporting Information S1) and assuming a uniform change over time, we suggest that a Vp change over 2 years or more and a Qp change over 9 months or more are detectable using DDV and DDQ tomography. The other major factors influencing the performance of tomography include the distribution of seismic stations and events and the quality of seismic data.

However, this provides only a crude estimate of the minimum period required to monitor the short-term medium change using the data from our long-term study, considering the short-term change could be of a different amplitude. For example, Hutchings et al. (2014), Jeanne et al. (2015), and Rutqvist et al. (2016) show that 4-D tomography can effectively detect significant medium changes after 1 and 2 months of injection around an EGS area of the northwest Geysers. Nevertheless, our uncertainty analysis indicates that at least 1% Vp change and 5% Qp change are required to be detectable using 4-D tomography. It is also worth noting that our results show the relative amplitude of the Qp change and Qp uncertainty is ~3 times larger than that for Vp, indicating that detecting Qp change is more effective than detecting Vp change.

6. Conclusions

We have developed high-resolution temporal change models of Vp and Qp for The Geysers geothermal field between the years 2005 and 2011 using DDV and DDQ tomography. To mitigate artifacts in the temporal change models resulting from the nonuniform data coverage of the two periods, we carefully selected closely located pairs of earthquakes for the two periods and common station observations for the selected earthquakes. Resolution tests show that the southeastern part of the northwest Geysers is well resolved from ~1 to ~4-km depth, covering both the NTR and HTR.

Our results show that beneath the injection wells with large injection volumes between 2005 and 2011, the NTR and HTR have different seismic responses to the injected water, which is likely related to the differences in their geological properties. The NTR and the upper part of the HTR are dominated by zones with decreased Vp and Qp and broadly distributed seismicity, indicating increased and diffuse fracturing and increased but still partial saturation. The lower part of the HTR shows localized zones with decreased Vp, increased Qp, and more concentrated or linearly distributed seismicity, which may indicate full fluid saturation within localized permeable fracture/fault zones due to continuous, confined fluid flow. The different patterns in fluid flow are consistent with the fact that the NTR, composed of hydrothermally altered and highly fractured metagraywacke, has high permeability whereas the lower part of the HTR, composed of granite, has lower permeability based on previous geological and core studies. In contrast to the low porosity and low permeability in the upper part of the HTR to the northwest of our resolved area composed of hornfelsic metagraywacke, our results suggest high porosity and

high permeability in the upper part of the HTR in our resolved area. The high permeability in the upper part of the HTR and the pre-existing permeable fault zones in the lower part of the HTR can account for the success of the inadvertently created EGS and thus high steam productivity in the southeastern part of the northwest Geysers.

Our results indicate that the joint analysis of temporal changes in velocity and attenuation as well as seismicity is valuable for monitoring reservoir condition changes related to fluid injection. Compared to a standard approach of imaging the velocity change, imaging the attenuation change is particularly helpful for characterizing the fluid saturation state of the reservoir. The greater sensitivity of attenuation to medium changes is evidenced by the larger amplitude of observed attenuation changes than velocity changes. The greater relative amplitude of the observed attenuation change and the associated uncertainty than that for velocity indicates that attenuation changes are more detectable. These suggest that attenuation tomography is a more effective way of detecting spatial and temporal reservoir medium changes than velocity tomography. Considering that the production from the HTR due to water injection in our resolved area is similar to the concept of EGS, our study also serves as a valuable example for future EGS projects at The Geysers and other geothermal systems.

Data Availability Statement

Waveform data, metadata, or data products for this study were accessed through the NCEDC, https://doi.10.7932/NCEDC. The list of earthquakes used for the 4-D tomography study with their relocations and Vp and Qp models are provided as supplementary files with a brief description in Supporting Information S2.

Acknowledgments
We thank Roland Gritto for providing us the initial lists of 2005 and 2011 events. We thank Craig Hartline for providing us the well data and for discussions on reservoir conditions. We thank the Editor, Associate Editor, and an anonymous reviewer for providing very helpful comments to improve our manuscript. This work was supported by the National Science Foundation under awards EAR-1724685 and EAR-2042919.

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